

EXPERIMENT 23. Text Coherence Evaluation

Aim: To develop a Python program that evaluates the coherence of a given text.

Program:

```
import spacy

nlp = spacy.load("en_core_web_sm")

text = input("Enter a text: ")

sentences = [sent.text for sent in nlp(text).sents]

vectors = []

for sentence in sentences:

    doc = nlp(sentence)

    vectors.append(doc.vector)

scores = []

for i in range(len(vectors) - 1):

    doc1 = nlp(sentences[i])

    doc2 = nlp(sentences[i + 1])

    similarity = doc1.similarity(doc2)

    scores.append(similarity)

if scores:

    average = sum(scores) / len(scores)

else:

    average = 1.0

print("\nCoherence Score:", round(average, 3))

if average >= 0.6:

    print("Text is highly coherent.")

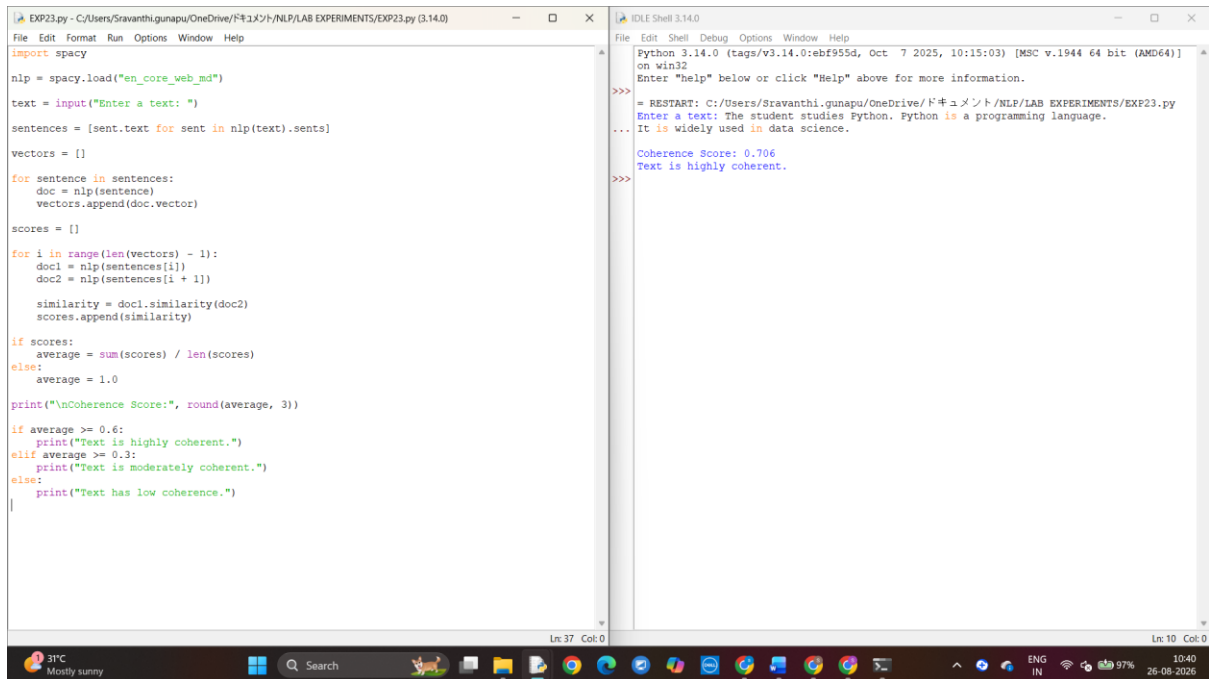
elif average >= 0.3:

    print("Text is moderately coherent.")

else:

    print("Text has low coherence.")
```

Output:



```
EXP23.py - C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP23.py (3.14.0)
File Edit Format Run Options Window Help

import spacy

nlp = spacy.load("en_core_web_md")

text = input("Enter a text: ")
sentences = [sent.text for sent in nlp(text).sents]

vectors = []

for sentence in sentences:
    doc = nlp(sentence)
    vectors.append(doc.vector)

scores = []

for i in range(len(vectors) - 1):
    doc1 = nlp(sentences[i])
    doc2 = nlp(sentences[i + 1])

    similarity = doc1.similarity(doc2)
    scores.append(similarity)

if scores:
    average = sum(scores) / len(scores)
else:
    average = 1.0

print("\nCoherence Score:", round(average, 3))

if average >= 0.6:
    print("Text is highly coherent.")
elif average >= 0.3:
    print("Text is moderately coherent.")
else:
    print("Text has low coherence.")
|

IDLE Shell 3.14.0
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP23.py
Enter a text: The student studies Python. Python is a programming language.
... It is widely used in data science.

Coherence Score: 0.706
Text is highly coherent.
>>>
```